



CHANDIGARH UNIVERSITY

Discover. Learn. Empower.

ASSIGNMENT - 2

Subject: Computer Networks

Subject Code: 24CST-339

Submitted by:

Name: Bharti Mehra

UID: 24BAI70051

Section: 24AIT_KRG-1_B

Submitted to:

Faculty Name: Alok Sir

Assignment – 02

Q1. What is an IP Address? Why do we need classes in case of IPV4 address? Explain each class. Mention how many host and networks are possible in each class. Also explain what is subnetting and supernetting with examples.

Ans. An IP Address (Internet Protocol Address) is a unique numerical label assigned to each device connected to a computer network that uses the Internet Protocol for communication. It serves two main purposes:

- Identifying a device on the network.
- Locating the device to enable communication with other devices over a network like the Internet.

An IP address has the following components:

1. Network Portion: Identifies the network to which the device belongs.
2. Host Portion: Identifies the individual device on the network.
3. Subnet Mask (for IPv4): Defines which part of the IP is network and which part is host.

IPv4 originally used a classful addressing system to divide the 32-bit address space into different classes according to the size of the network. Different organizations require different numbers of IP addresses. For example:

- A very large organization may need thousands of host addresses.
- A medium-sized organization may need hundreds.
- A small organization may need only a few addresses.

Therefore, IPv4 addresses were divided into five classes: A, B, C, D and E, so that addresses could be allocated according to different requirements.

Class A

Range: 1.0.0.0 to 126.255.255.255

Default subnet mask: 255.0.0.0 or /8

Network bits: 8

Host bits: 24

Number of networks: $2^7 = 126$

0 and 127 are reserved, leaving 126 usable addresses.

Hosts per network: $2^{24} - 2 = 16,777,214$

Class B

Range: 128.0.0.0 to 191.255.255.255

Default subnet mask: 255.255.0.0 or /16

Network bits: 16

Host bits: 16

Number of networks: $2^{14} = 16,384$

Hosts per network: $2^{16} - 2 = 65,534$

Class C

Range: 192.0.0.0 to 223.255.255.255

Default subnet mask: 255.255.255.0 or /24

Network bits: 24

Host bits: 8

Number of networks: $2^{21} = 2,097,152$

Hosts per network: $2^8 - 2 = 254$

Class D

Range: 224.0.0.0 to 239.255.255.255

Class D addresses are used for multicasting.

They do not have the traditional network/host division.

Class E

Range: 240.0.0.0 to 255.255.255.255

Class E addresses are reserved for experimental and research purposes.

They are not normally used for assigning addresses to ordinary hosts.

Subnetting is the process of dividing a large network into smaller logical networks called subnets. It is done by borrowing bits from the host portion and using them as subnet bits.

Example:

Suppose we have: 192.168.1.0/24

A /24 network has:

- 24 network bits
- 8 host bits
- 256 total addresses
- 254 usable host addresses

If we subnet it into 4 subnets, we borrow 2 host bits:

$$2^2 = 4 \text{ subnets}$$

The new prefix becomes: /26

Each subnet has:

$$2^6 = 64 \text{ total addresses}$$

and: $64 - 2 = 62$ usable host addresses.

The four subnets are:

1. 192.168.1.0/26
2. 192.168.1.64/26
3. 192.168.1.128/26
4. 192.168.1.192/26

Supernetting is the opposite of subnetting. It is the process of combining multiple smaller networks into one larger logical network. It is also called route aggregation or route summarization. In supernetting, bits are borrowed from the network portion and added to the host portion, resulting in a shorter prefix.

Example:

Suppose we have four consecutive Class C networks:

192.168.0.0/24

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

These four networks can be combined into: 192.168.0.0/22

A /22 network contains:

$$2^{32-22} = 2^{10} = 1024$$

total addresses, or 1022 usable host addresses under traditional IPv4 host-address rules.

The combined range is: 192.168.0.0 – 192.168.3.255

Q2. What is the difference between Link state and Distance vector routing? Explain how these two algorithms help in identifying the least cost path with the help of examples.

Ans. Routing algorithms are used by routers to determine the best or least-cost path for forwarding data from a source network to a destination network. Two important routing approaches are Link State Routing and Distance Vector Routing.

Link State Routing

In Link State Routing, every router builds a complete map (or topology) of the network. Each router collects information about its directly connected links and shares this information with other routers. After obtaining the network topology, each router independently calculates the least-cost path using Dijkstra's Shortest Path Algorithm.

Steps:

1. Each router discovers its directly connected neighbors.
2. It determines the cost of each connected link.
3. It creates a Link State Advertisement (LSA) containing this information.
4. LSAs are flooded throughout the network.
5. Every router builds the same network topology database.
6. Each router runs Dijkstra's algorithm to find the least-cost path.

Consider the following network:

A---B = 2

B---D = 1

C---D = 2

A---C = 5

To reach D from A, we have two paths:

- A – B – D at cost 3
- A – C – D at cost 7

Therefore, least cost path is A – B – D.

Distance Vector Routing

In Distance Vector Routing, a router does not have the complete network topology. Instead, it knows:

- The cost to its directly connected neighbors.
- The distance (cost) to destinations through those neighbors.

Routers periodically exchange their routing tables with their neighboring routers. The algorithm uses the Bellman-Ford algorithm to calculate the least-cost path.

Steps:

1. Each router initially knows the cost to its directly connected neighbors.
2. It maintains a routing table containing the best-known distance to each destination.
3. Routers exchange their distance vectors with neighboring routers.
4. A router compares different paths through its neighbors.
5. It selects the path having the minimum total cost.
6. The information is repeatedly exchanged until the routing tables converge.

Consider the following network:

A---B = 2

B---D = 1

C---D = 2

A---C = 5

Suppose router A wants to find the least-cost path to D.

A knows:

Cost from A to B = 2

Cost from A to C = 5

Suppose A receives from B that B can reach D with a cost of 1.

A calculates:

$$\begin{aligned}\text{Cost (A - B - D)} &= \text{Cost (A, B)} + \text{Cost (B, D)} \\ &= 2 + 1 = 3\end{aligned}$$

Through C:

$$\text{Cost (A - C - D)} = \text{Cost (A, C)} + \text{Cost (C, D)}$$

$$= 5 + 2 = 7$$

Therefore, A selects:

$A \rightarrow B \rightarrow D$ with a minimum cost of 3.

Differences between the two are:

Feature	Link State Routing	Distance Vector Routing
Network knowledge	Complete network topology	Information about destinations through neighbours
Information exchanged	Link-state information	Routing table/distance vector
Algorithm	Dijkstra's algorithm	Bellman-Ford algorithm
Updates	Generally event-driven/flooded when topology changes	Periodic exchange with neighbours
Calculation	Each router calculates shortest paths from topology	Routers calculate distances through neighbours
Convergence	Generally faster	Generally slower
Complexity	More complex	Less Complex

Q3. What is congestion in a network? Why do we need to control congestion? Explain the difference between Leaky bucket and Token bucket with examples. Also, explain why do we need to do fragmentation?

Ans. Network congestion occurs when the amount of data entering a network is greater than the network's capacity to process and forward it. When too many packets arrive at routers or links simultaneously, the available bandwidth and buffer space may become insufficient. As a result, packets may experience delays or may even be dropped.

Example: Suppose a network link can transmit 10 Mbps, but devices are sending data at 20 Mbps. Since the link can handle only 10 Mbps, packets start accumulating

in the router's buffer. If the buffer becomes full, packets may be dropped, causing retransmissions and further congestion.

Congestion control is needed to prevent excessive traffic from overwhelming the network.

It helps to:

- Reduce packet loss caused by overflowing router buffers.
- Reduce network delay and waiting time.
- Improve network throughput by using available bandwidth efficiently.
- Prevent congestion collapse, where increasing traffic can actually reduce useful data delivery.
- Provide better network performance and reliability.

Leaky Bucket Algorithm

The Leaky Bucket algorithm is a congestion-control and traffic-shaping technique that controls the rate at which packets are transmitted into the network. It works like a bucket with a small hole at the bottom. Packets can enter the bucket at varying rates, but they are transmitted at a fixed rate.

Example:

Bucket capacity = 5 packets

Output rate = 1 packet/second

If 5 packets arrive suddenly, they are stored in the bucket and transmitted one at a time:

Time	Packets transmitted
1 sec	1
2 sec	1
3 sec	1
4 sec	1
5 sec	1

Thus, even though packets arrived in a burst, the network receives them at a constant rate. If more than 5 packets arrive while the bucket is full, the additional packets are discarded.

Token Bucket Algorithm

The Token Bucket algorithm also controls traffic, but it is more flexible because it allows bursts of traffic.

In this algorithm, tokens are generated and placed into a bucket at a fixed rate.

- One token generally permits one packet (or a specified amount of data) to be transmitted.
- Tokens accumulate when there is no traffic, up to the bucket capacity.
- When packets arrive, they use the available tokens.
- If enough tokens are available, packets can be transmitted as a burst.

Example:

Bucket capacity = 5 tokens

Token generation rate = 1 token/second

If the bucket has accumulated 5 tokens and 5 packets arrive together, all 5 packets can be transmitted as a burst, because there are 5 tokens available. After that, if no tokens are left, packets must wait for new tokens.

Fragmentation is the process of dividing a large IP packet into smaller fragments so that it can travel through a network whose link has a smaller Maximum Transmission Unit (MTU).

Different networks can have different maximum packet sizes. If the 4000-byte packet is larger than the next network's MTU, it cannot be transmitted as a single packet. Therefore, it may be divided into smaller fragments.

Example: Suppose an IP packet is 4000 bytes and the next network has an MTU of 1500 bytes. The packet can be divided into smaller fragments, approximately:

Fragment 1 $\leq$ 1500 bytes

Fragment 2 $\leq$ 1500 bytes

Fragment 3 = Remaining